

WEIGHTED STOKES AND ORSTEIN-UHLENBECK DIFFUSION

NOTES AND AREAS FOR FUTURE EXPLORATION.

Charbel A. Dargham

I- INTRODUCTION:

Our motif for writing this is to give ourselves motivation to learn fundamental concepts and explore rudimentary areas of somewhat modern mathematics. The goal of this paper is to build one rigorous chain and then validate it computationally

$$g \rightarrow dvol_g \rightarrow d\mu = e^{-V} dvol_g \rightarrow div_\mu \rightarrow \Delta_v = div_\mu \nabla \rightarrow \Delta_\gamma = \Delta - x \cdot \nabla \rightarrow \partial_t u = \Delta_\gamma u$$

Weighted manifolds – also called manifolds with density or smooth metric measure spaces – are standard objects in modern geometry and analysis, and they arise naturally in diffusion theory, Bakry – Emery geometry, Ricci flow, Sobolev inequalities, and related weighted Laplacian problems.

We start by understanding what Smooth manifolds and k-forms are.

II- REVIEW MATERIAL: Smooth manifolds and forms:

For purposes of brevity, we'll only be writing summaries of what is necessary to write this paper.

- A smooth n-manifold is a Hausdorff and, for our purposes, second countable topological space M that $\forall p \in M \exists U \in \mathcal{N}(p); U \cong \mathbb{R}^n$
- Let M be a topological n-manifold. A coordinate chart on M is a pair (U, φ) , where U is an open and $\varphi: U \rightarrow \hat{U} \subset \mathbb{R}^n$ is a homeomorphism from U to an open subset $\hat{U}$ of $\mathbb{R}^n$.
- An atlas is a collection of charts covering M, the atlas is smooth whenever two charts overlap,

$$(U, \varphi) \quad (V, \psi)$$

The transition map

$$\psi \circ \varphi^{-1}: \varphi(U \cap V) \rightarrow \psi(U \cap V)$$

Is smooth.

- Let M be a smooth manifold

A function

$$f: M \rightarrow \mathbb{R}$$

Is smooth if, in every chart

$$f \circ \varphi^{-1}: \varphi(U) \subset \mathbb{R}^n \rightarrow \mathbb{R}$$

Is smooth in the usual multivariable calculus sense.

More generally, if

$$F: M \rightarrow N$$

Is a map between manifolds and F is smooth if

$$\psi \circ F \circ \varphi^{-1}$$

Is a smooth map between open subsets of Euclidean spaces.

For tangent vectors

A tangent vector is an operator that differentiates functions

More precisely

Suppose that (U, φ) is a chart, $\forall p \in U$

$$\varphi(p) = \left(x^i(p) \right)_{i \in \{1, \dots, n\}}$$

Then a tangent vector at p, $v \in T_p M$, would be something of the form

$$v = (v^1, \dots, v^n) \cdot \left(\frac{\partial}{\partial x^i} \right) \Big|_p = \sum_{i=1}^n v_i \frac{\partial}{\partial x^i} \Big|_p$$

For example, if we have a regular surface in $\mathbb{R}^3$, parametrized by

$$\sigma(u, v) = (x^1(u, v), x^2(u, v), x^3(u, v))$$

Then a vector in the tangent space $X \in T_p \Sigma$, $p = \sigma(u_0, v_0)$ would look like

$$X = v_1 \sigma_u(u_0, v_0) + v_2 \sigma_v(u_0, v_0)$$

More precisely $T_p \Sigma = \text{span}\{\sigma_u, \sigma_v\} \Big|_p$

The Cotangent space at p is the dual vector space of $T_p^* M = \text{Hom}(T_p M, \mathbb{R})$

They're what gives us forms, differential forms that is precisely:

recall that the basis of $T_p M = \left(\frac{\partial}{\partial x^i}\right)_{1 \leq i \leq n}$ then the dual basis of T_p^* would be

$$T_p^* M = \langle dx^i \rangle_{1 \leq i \leq n}$$

That satisfy $dx^i \left(\frac{\partial}{\partial x^j}\right) = \delta_j^i$

$$v = \sum_{j=1}^n \frac{v^j \partial}{\partial x^j} \Rightarrow dx^i(v) = v^i$$

Recall from multivariate calculus that the differential of a 0-form creates a 1-form, and basically this is what df entails if $f: M \rightarrow \mathbb{R}$ is smooth, then the differential of f at p is the covector

$$df_p: T_p M \rightarrow \mathbb{R}$$

$$df_p(v) = v(f)$$

So df_p eats a tangent vector and returns the directional derivative of f in that direction

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x^i} dx^i$$

In Euclidean space, people identify df with ∇f , but this secretly uses the "*Euclidean metric*"

Without a metric, df still exists, this distinction will matter later when we define

$$\Delta_\mu f = \text{div}_\mu(\nabla f)$$

A nice example would be the following

Let

$$H_N = \text{span}\{e_1, \dots, e_N\} \subset L^2(0, \pi)$$

Where

$$e_k(x) = \sqrt{\frac{2}{\pi}} \sin(kx)$$

Every element $u \in H_N$ has the form

$$u = a_1 e_1 + \cdots + a_N e_N$$

We define

$$M_N = \{u \in H_N : \|u\|_{L^2} = 1\}$$

Define

$$\Phi: H_N \rightarrow \mathbb{R}, \quad \Phi(u) = \|u\|_{L^2}^2$$

Then

$$M_N = \Phi^{-1}(1).$$

The derivative of Φ at u in the direction of $h \in H_N$ is

$$D\Phi(u)(h) = 2(u, v)_{L^2}$$

If $u \in M_N$, then $u \neq 0 \Rightarrow D\Phi_u \neq 0$, then 1 is a regular value of Φ , and M_N is a smooth $(N - 1) - manifold$.

$$T_u M_N = \{v \in H_N : v \in \ker D\phi_u\} = \{v \in H_N : v \in \langle u, v \rangle_{L^2} = 0\}$$

Now consider the functional

$$F: M_n \rightarrow \mathbb{R}$$

$$F(u) = \int_0^\pi q(x) u(x)^2 dx$$

Then

$$dF_u(v) = 2 \int_0^\pi q(x) u(x) v(x) dx$$

Observe, we are working in

$$H_N = span\{e_1, \dots, e_N\}$$

So, every $u \in H_N$ is

$$u = \sum_{k=1}^N a_k e_k$$

$$u \rightarrow (a^1(u), \dots, a^N(u))$$

So that our coordinate chart is $a^i(u) = \langle u, e_i \rangle_{L^2}$

And hence our $T_u^* M_N = \text{span}\{da_1, \dots, da_n\}$

$$\begin{aligned} 2 \int_0^\pi q(x) u(x) v(x) dx &= 2 \int_0^\pi q(x) \sum_i a_i e_i(x) \sum_j v^j e_j(x) dx \\ &= 2 \sum_{i,j} a_i v^j \left[\int_0^\pi q(x) e_i e_j dx \right] \\ &= 2 \sum_{i,j} a_i Q_{ij} v^j = a^t [Q] v \end{aligned}$$

Notice how $\frac{\partial F}{\partial a_j} = 2 \sum_i a_i Q_{ij}$, then

$$dF = a^t [Q] = \sum_{j=1}^N 2 \frac{\partial F}{\partial a_j} da_j$$

At a point $p \in M$, a k - *form* is an alternating multilinear map

$$\omega_p: \prod_{1 \leq i \leq n} T_p M \rightarrow \mathbb{R}$$

It takes k tangent vectors as input and returns a number. Geometrically:

A 1-form measures oriented length along a direction.

A 2-form measures oriented area... a 3-form measures oriented volume...

In local coordinates

$$(x^1, \dots, x^n),$$

a 1 - *form* looks like

$$\alpha = \sum_{i=1}^n a_i dx^i$$

A n-form looks

$$\omega = \sum_{i_1 < \dots < i_k} w_{i_1 \dots i_k} \bigwedge_{i=1}^k dx^{i_i}$$

The space of smooth k-forms on M is denoted

$$\Omega^k(M).$$

The wedge product combines forms, if $\alpha \in \Omega^k(M), \beta \in \Omega^\ell(M)$ then

$$\alpha \wedge \beta \in \Omega^{k+\ell}(M)$$

If $\dim M = n$, then to integrate over all of M, you need something that measures n -dimensional volume, which is precisely an n -form.

On a Riemannian manifold, the metric gives a natural volume form

$$dvol_g = \sqrt{\det(g_{ij})} dx^1 \wedge \dots \wedge dx^n$$

So we can integrate

$$\int_M f dvol_g$$

Suppose $F: N \rightarrow M$ is a smooth map, if $\omega \in \Omega^k(M)$, it's pullback is a k-form

$$F^* \omega \in \Omega^k(N)$$

At a point $p \in N$, we define

$$(F^* \omega)_p(v_1, \dots, v_k) = \omega_{F(p)}(dF_p(v_1), \dots, dF_p(v_k))$$

Pullback lets us integrate forms using parametrizations if

$$\Phi: U \subset \mathbb{R}^k \rightarrow M$$

Parametrizes a k-dimensional piece of M, then

$$\int_{\Phi(U)} \omega = \int_U \Phi^* \omega$$

So pullback is the rigorous version of substitution

Examples:

Take the Finite Fourier space

$$H_N = \text{span}_{\mathbb{C}}\{e_{-N}, \dots, e_N\} \subset L^2(S^1; \mathbb{C}),$$

Where

$$e_k(x) = \frac{1}{\sqrt{2\pi}} e^{ikx}.$$

So, every function is

$$u(x) = \sum_{k=-N}^N z_k e_k(x), \quad z_k \in \mathbb{C}$$

Hence $H_N \cong \mathbb{C}^{2N+1}$, Now impose the L^2 – *unit* constraint:

$$\mathcal{S}_N = \{u \in H_N : \|u\|_{L^2} = 1\}$$

we'll show that it's a **real** manifold, it important to distinguish it from it being a complex manifold, which is a totally different thing, (the defining equation must be independent of $\bar{z}$ *for holomorphicity*)

because the Fourier basis is orthonormal

$$\|u\|_{L^2}^2 = \sum_{n=-N}^N |z_k|^2$$

So defining

$$\Phi: \mathbb{C}^m \rightarrow \mathbb{R}, \quad \Phi(z) = \sum_{k=-N}^N |z_k|^2$$

Write $z_k = a_k + ib_k$, $\Phi(z) = \sum_{k=-N}^N a_k^2 + b_k^2$

$$d\Phi_z(\zeta) = 2\operatorname{Re}\langle z, \zeta \rangle_{\mathbb{C}^m}$$

For $z \in S_N$, $d\Phi_z \neq 0_{\operatorname{Hom}(H_N, \mathbb{R})}$, making it a submersion and giving us a smooth real submanifold of $\mathbb{C}^m$ of codimension 1.

Thus $\mathcal{S}_N$ is a smooth manifold of real dimension, with $T_z S_N = \{\zeta \in \mathbb{C}^m : \operatorname{Re}\langle z, \zeta \rangle = 0\}$

Writing the 1-forms, each complex coordinate is

$$z_k = a_k + ib_k$$

So the real coordinate 1-forms are da_k & db_k

With

$$dz_k = da_k + idb_k$$

$$d\bar{z}_k = da_k - idb_k$$

A basic Kahler-style 2-form on $\mathbb{C}^m$ is

$$\omega_0 = \frac{i}{2} \sum_k dz_k \wedge d\bar{z}_k$$

But $dz_k \wedge d\bar{z}_k = (da_k + idb_k) \wedge (da_k - idb_k) = -2ida_k \wedge db_k$

$$\omega_0 = \sum_k da_k \wedge db_k$$

Now consider the inclusion:

$$\iota: S_N \hookrightarrow \mathbb{C}^m$$

What the pullback $\iota^*\omega_0$,

For a smooth map $f: M \rightarrow N$, between manifolds and a k -form ω , the pullback $f^*\omega$ is the k -form on M defined pointwise by

$$(f^*\omega)_p(v_1, \dots, v_k) = \omega_{f(p)}(df_p(v_i))$$

Computationally

$$(\iota^*\omega_0)_p(u, v) = \omega_0|_p(u, v) = u, v \in T_p S_N$$

It remains the same, but we only evaluate it on vectors that are tangent to the sphere, what changes is the domain on which we're allowed to evaluate it.

This is the general principle of pulling back along an embedding: a pulled-back form is the original form restricted to the tangent directions of the submanifold.

On all of $\mathbb{C}^m$, ω_0 is nondegenerate, however pulling back to the sphere, this is no longer the case, there is a tangent direction along which ω_0 vanishes against everything tangent. That direction is exactly what the image is identifying.

The phase action. The circle group $S^1 = \{e^{i\theta}\}$ acts on $\mathbb{C}^m$ by

$$\theta \cdot z = e^{i\theta} z$$

This action preserves the norm, so the vector remains on the sphere.

The vector field generating the phase rotation is $z \rightarrow iz$, so $iz \in T_z \mathcal{S}_N$, this direction has a name: the phase direction, or the Hopf direction.

$$(\iota^* \omega_0)_z(iz, v) = \text{Im}\langle iz, v \rangle = -\text{Re}\langle z, v \rangle = 0$$

So iz is in the kernel of $\iota^* \omega_0$ at every point. The pulled-back form is **degenerate**, with a 1-dimensional kernel given exactly by the phase direction $\mathbb{R} \cdot iz$. This is no accident: it's a general fact that pulling a symplectic form back to an odd-dimensional submanifold must produce a degenerate form (symplectic forms only exist in even dimensions), and the kernel is called the *characteristic foliation*.

The quotient: complex projective space

Because the phase direction is the kernel of $\iota^* \omega_0$, if you collapse it (i.e., quotient the sphere by the phase S^1 -action), the form descends to a *nondegenerate* 2-form on the quotient. The quotient is

$$\mathcal{S}_N/S^1 = \mathbb{CP}^{m-1},$$

complex projective space of complex dimension $m - 1$. This is the **Fubini–Study symplectic manifold**: there is a unique 2-form ω_{FS} on $\mathbb{CP}^{m-1}$ such that $\pi^* \omega_{FS} = \iota^* \omega_0$, where $\pi: \mathcal{S}_N \rightarrow \mathbb{CP}^{m-1}$ is the quotient projection. ω_{FS} is nondegenerate, closed, and is in fact a Kähler form. This whole construction is the simplest nontrivial case of **symplectic reduction** (Marsden–Weinstein), with the moment map being $\mu(z) = \frac{1}{2} \|z\|^2$ for the S^1 -action.

Start concrete. $\mathbb{C}^m$ is a complex vector space of dimension m . Throw away the origin, then declare two nonzero vectors **equivalent** if one is a complex scalar multiple of the other:

$$z \sim z' \iff z' = \lambda z \text{ for some } \lambda \in \mathbb{C}^*.$$

The set of equivalence classes is **complex projective space** $\mathbb{CP}^{m-1}$. A point of $\mathbb{CP}^{m-1}$ is therefore a *complex line through the origin* in $\mathbb{C}^m$. We write the equivalence class of $(z_0, z_1, \dots, z_{m-1})$ as

$$[z_0 : z_1 : \cdots : z_{m-1}],$$

with colons reminding you that only ratios matter.

There are two equivalent ways to package this:

- **Lines through 0 picture.** Each point of $\mathbb{CP}^{m-1}$ is a complex line.
- **Sphere quotient picture.** Each line meets the unit sphere $S^{2m-1} \subset \mathbb{C}^m$ in a circle (multiply by any $e^{i\theta}$ and you stay on the line and on the sphere). So

$$\mathbb{CP}^{m-1} = S^{2m-1}/S^1,$$

which is the perspective of the previous discussion.

Both pictures will appear: the chart construction below uses the "lines" view, and the section **s** picks out a unit-norm representative on each line, connecting back to the sphere.

Manifolds are studied through **charts**: local coordinate systems that identify a piece of the manifold with an open subset of Euclidean (or in this case complex) space. To do calculus on $\mathbb{CP}^{m-1}$, we need such charts.

The standard atlas. For each $k = 0, 1, \dots, m-1$, define the open set

$$U_k = \{[z_0 : \cdots : z_{m-1}] \in \mathbb{CP}^{m-1} : z_k \neq 0\}.$$

These cover $\mathbb{CP}^{m-1}$ (every point has at least one nonzero coordinate). On U_k , since $z_k \neq 0$, we can rescale the representative by dividing through by z_k :

$$[z_0 : \cdots : z_{m-1}] = \left[\frac{z_0}{z_k} : \cdots : 1 : \cdots : \frac{z_{m-1}}{z_k} \right]$$

(the 1 is in the k -th slot). The remaining $m-1$ ratios are well-defined complex numbers and give a bijection $U_k \rightarrow \mathbb{C}^{m-1}$.

The chart on U_0 — the one in the image. Take $k = 0$, so we look at points where $z_0 \neq 0$.

Define

$$w_j = \frac{z_j}{z_0}, j = 1, \dots, m-1.$$

The map $[z_0:z_1:\dots:z_{m-1}] \mapsto (w_1, \dots, w_{m-1})$ is a homeomorphism (in fact biholomorphism) $U_0 \rightarrow \mathbb{C}^{m-1}$. The w_j are **local complex coordinates** on $\mathbb{CP}^{m-1}$ near any point with $z_0 \neq 0$.

This is the analog of looking at the affine plane $z_0 = 1$ inside $\mathbb{C}^m$: every line through the origin not parallel to that plane meets it in exactly one point, and that point's last $m-1$ coordinates are the w_j .

Mental picture. Just as on the round sphere we use stereographic projection to flatten a piece of sphere into a plane and do calculus there, here we flatten a piece of projective space into $\mathbb{C}^{m-1}$ using the ratios w_j . The form ω_{FS} lives on $\mathbb{CP}^{m-1}$; to write it down concretely we'll express it in these flattened coordinates.

Now comes a beautiful trick. We have:

- The sphere $\mathcal{S}_N \subset \mathbb{C}^m$, with the pulled-back form $\iota^* \omega_0$.
- The projection $\pi: \mathcal{S}_N \rightarrow \mathbb{CP}^{m-1}$, $z \mapsto [z]$.
- The quotient form ω_{FS} on $\mathbb{CP}^{m-1}$, defined by $\pi^* \omega_{FS} = \iota^* \omega_0$.

We want to write ω_{FS} in our coordinates $(w_1, \dots, w_{m-1})$ on U_0 . The cleanest way: find a section of π over U_0 — that is, a smooth map

$$s: \mathbb{C}^{m-1} \rightarrow \mathcal{S}_N \subset \mathbb{C}^m$$

Constructing s : Start with the line corresponding to $w = (w_1, \dots, w_{m-1})$. A natural representative on this line, with $z_0 = 1$, is

$$(1, w_1, \dots, w_{m-1}) \in \mathbb{C}^m.$$

This vector has squared norm

$$\| (1, w_1, \dots, w_{m-1}) \|^2 = 1 + \sum_{j=1}^{m-1} |w_j|^2 = 1 + \|w\|^2.$$

To get a unit vector on the same line, divide by the norm:

$$s(w) = \frac{1}{\sqrt{1 + \|w\|^2}} (1, w_1, \dots, w_{m-1}).$$

That's the formula in the image. Note it is *not* the only choice — we could multiply by any phase $e^{i\varphi(w)}$ and still get a section. Different sections give different liftings, but the *pulled-back form* $s^*\omega_0$ will be the same regardless, because phase shifts move you along the kernel direction $i\mathbf{z}$ of $\iota^*\omega_0$. This is the "gauge invariance" that makes the construction well-defined.

Why use a section instead of working on $\mathbb{CP}^{m-1}$ directly? Because forms on $\mathbb{C}^m$ are easy to write — we have global coordinates $z_0, \dots, z_{m-1}$. Forms on $\mathbb{CP}^{m-1}$ are awkward to write directly. So we lift back to $\mathbb{C}^m$, compute there, and pull the answer down. The section s is the lift.

The relationship between $s^*\omega_0$, $\iota^*\omega_0$, and ω_{FS} is:

$$s^*\omega_0 = s^*(\iota^*\omega_0) \text{ (since } s \text{ lands in } \mathcal{S}_N) = (\iota \circ s)^*\omega_0,$$

and using $\pi \circ s = \text{id}_{U_0}$ and $\pi^*\omega_{FS} = \iota^*\omega_0$:

$$s^*(\iota^*\omega_0) = s^*\pi^*\omega_{FS} = (\pi \circ s)^*\omega_{FS} = \omega_{FS}|_{U_0}.$$

So computing $s^*\omega_0$ is computing ω_{FS} in our chart. That's the whole strategy.

If you have a 2-form on $\mathbb{C}^m$ written as

$$\alpha = \sum f_{kl}(z) dz_k \wedge d\bar{z}_l$$

and a smooth map $s: \mathbb{C}^{m-1} \rightarrow \mathbb{C}^m$ given by component functions $z_k = s_k(w)$, then $s^*\alpha$ is computed by the algorithm:

1. Replace each function $f_{kl}(z)$ by $f_{kl}(s(w))$.
2. Replace each dz_k by $\left(\sum_j \frac{\partial s_k}{\partial w_j} dw_j\right)$ (and similarly $d\bar{z}_l$ by $\left(\sum_j \frac{\partial \bar{s}_l}{\partial \bar{w}_j} d\bar{w}_j\right)$)
3. Wedge everything out, collect terms.

On a complex manifold the exterior derivative splits as

$d = \partial + \bar{\partial}$, where:

- $\partial f = \sum_j \frac{\partial f}{\partial z_j} dz_j$ is the holomorphic part.
- $\bar{\partial} f = \sum_j \frac{\partial f}{\partial \bar{z}_j} d\bar{z}_j$ is the antiholomorphic part.

Two key identities:

- $\partial^2 = 0$, $\bar{\partial}^2 = 0$, and $\partial \bar{\partial} = -\bar{\partial} \partial$.
- For any smooth real-valued function K , the 2-form $i \partial \bar{\partial} K$ is automatically real and closed. (Closed because $d(\partial \bar{\partial} K) = (\partial + \bar{\partial}) \partial \bar{\partial} K = 0$ using $\partial^2 = \bar{\partial}^2 = 0$.)

A function K such that the symplectic/Kähler form is $\omega = i \partial \bar{\partial} K$ is called a **Kähler potential**. The standard form on $\mathbb{C}^m$ has Kähler potential $K(z) = \frac{1}{2} \|z\|^2$ since

$$i \partial \bar{\partial} \left(\frac{1}{2} \sum_k z_k \bar{z}_k \right) = \frac{i}{2} \sum_k dz_k \wedge d\bar{z}_k = \omega_0.$$

The famous claim: On the chart U_0 of $\mathbb{CP}^{m-1}$,

$$\omega_{FS} = i \partial \bar{\partial} \log(1 + \|w\|^2),$$

where $\|w\|^2 = \sum_{j=1}^{m-1} |w_j|^2$, the idea is that we must introduce the logarithm as is we naively replaces $z = s(w)$ in the *kähler* form

$$\omega_0 = \frac{1}{2} \partial \bar{\partial} \left(\frac{1}{2} \|s(w)\|^2 \right)$$

Then nothing but zero will be shown, we want a construction that behaves nicely when we replace z by λz , Look at $\log \|z\|^2$

If we rescale $z \rightarrow \lambda z$ then

$$\log \|\lambda z\|^2 = \log(|\lambda|^2 \|z\|^2) = \log |\lambda|^2 + \log \|z\|^2$$

But the constant part vanishes upon differentiation.

Let us derive this from

$$\omega_{FS} = i \partial \bar{\partial} \log(1 + \|w\|^2)$$

Let $f(w, \bar{w}) = 1 + \sum_k w_k \bar{w}_k$. We want $i \partial \bar{\partial} \log f$.

Step 1: $\bar{\partial} \log f$. Compute:

$$\frac{\partial}{\partial \bar{w}_\ell} \log f = \frac{1}{f} \cdot \frac{\partial f}{\partial \bar{w}_\ell} = \frac{w_\ell}{f}.$$

So

$$\bar{\partial} \log f = \sum_\ell \frac{w_\ell}{f} d\bar{w}_\ell.$$

Apply ∂ to the above. $\partial g = \sum_j \frac{\partial g}{\partial w_j} dw_j$, and ∂ acts on the coefficient and treats $d\bar{w}_\ell$ as a constant 1-form (we then wedge dw_j on the *left*).

For each ℓ :

$$\frac{\partial}{\partial w_j} \left(\frac{w_\ell}{f} \right) = \frac{\delta_{j\ell}}{f} - \frac{w_\ell}{f^2} \cdot \frac{\partial f}{\partial w_j} = \frac{\delta_{j\ell}}{f} - \frac{w_\ell \bar{w}_j}{f^2}.$$

Therefore

$$\partial \bar{\partial} \log f = \sum_{j,\ell} \left(\frac{\delta_{j\ell}}{f} - \frac{w_\ell \bar{w}_j}{f^2} \right) dw_j \wedge d\bar{w}_\ell.$$

Combine the fractions over the common denominator f^2 :

$$\frac{\delta_{j\ell}}{f} - \frac{w_\ell \bar{w}_j}{f^2} = \frac{f \delta_{j\ell} - \bar{w}_j w_\ell}{f^2} = \frac{(1 + \|w\|^2) \delta_{j\ell} - \bar{w}_j w_\ell}{(1 + \|w\|^2)^2}.$$

Step 3: Multiply by i .

$$\omega_{FS} = i \sum_{j,\ell} \frac{(1 + \|w\|^2) \delta_{j\ell} - \bar{w}_j w_\ell}{(1 + \|w\|^2)^2} dw_j \wedge d\bar{w}_\ell.$$

The coefficient matrix $g_{j\bar{\ell}} = \frac{(1 + \|w\|^2) \delta_{j\ell} - \bar{w}_j w_\ell}{(1 + \|w\|^2)^2}$ is the **Fubini-Study Hermitian metric**.

It's positive definite (you can check), making $\mathbb{CP}^{m-1}$ into a Kähler manifold. The " $-\bar{w}_j w_\ell$ " piece is the rank-one correction that captures the fact that radial directions in w -space have already been quotiented out.

let us see how the integration of this two form would be played:

important caveat: this is a two form and $\mathbb{CP}^{m-1}$ has real dimension $2(m-1)$.

So, we require that $m=2$.

For $\mathbb{CP}^1$, the chart coordinate is just one complex number:

$$w \in \mathbb{C}$$

So

$$\begin{aligned} \omega_{FS} &= \frac{idw \wedge d\bar{w}}{(1 + |w|^2)^2} \\ \int_{\mathbb{CP}^1} \omega_{FS} &= \int_{\mathbb{C}} \frac{idw \wedge d\bar{w}}{(1 + |w|^2)^2} = \int_{\mathbb{R}^2} \frac{2}{(1 + x^2 + y^2)^2} dx dy = 2\pi \end{aligned}$$

The exterior derivative is an operator

$$d: \Omega^k(M) \rightarrow \Omega^{k+1}(M)$$

It generalizes gradient, curl, and divergence. It satisfies key properties On functions $f \in \Omega^0(M) = C^\infty(M)$

$$df = \sum_i \frac{\partial}{\partial x^i} dx^i$$

It satisfies the graded product rule

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta$$

And it squares to zero ($d^2 = 0$)

In coordinates if

$$\begin{aligned}\omega &= \sum_I \omega_I dx^{i_1} \wedge \dots \wedge dx^{i_k} \\ d\omega &= \sum_I d\omega_I dx^{i_1} \wedge \dots \wedge dx^{i_k}, \quad d\omega_I = \sum_j \frac{\partial \omega_I}{\partial x^j} dx^j \wedge dx^{i_1} \wedge \dots \wedge dx^{i_k} \\ d\omega &= \sum_{I,j} \frac{\partial \omega_I}{\partial x^j} dx^j \wedge dx^{i_1} \wedge \dots \wedge dx^{i_k}\end{aligned}$$

One important identity is that pullback commutes with the exterior derivative

$$d(F^*\omega) = F^*(d\omega)$$

We'll be using this fact to compute $d\omega$ before or after changing coordinates without worrying about consistency.

Let M be an oriented n -manifold. Suppose $\omega \in \Omega^n(M)$. In a positively oriented coordinate chart,

$$\omega = f(x^1, \dots, x^n) dx^1 \wedge \dots \wedge dx^n$$

Then locally we define

$$\int_U \omega = \int_{\varphi(U)} f(x^1, \dots, x^n) dx^1 \dots dx^n$$

This is coordinate-independent because under a change of variables, the top form transforms by the determinant of the Jacobian.

If the coordinate change is orientation-preserving, the determinant is positive, for a general manifold, one uses a partition of unity to glue the local definitions together.

For a smooth function f , df is a one form. The gradient ∇f is defined by

$$g(\nabla f, X) = df(X)$$

For all vector fields X . *df exists without a metric, but ∇f requires a metric.* The Laplacian $\Delta f = \text{div}(\nabla f)$ requires metric.

The generalized Stokes theorem says:

$$\int_M d\alpha = \int_{\partial M} \alpha$$

Here M is an oriented n -manifold with boundary, and

$$\alpha \in \Omega^{n-1}(M)$$

A manifold M is **orientable** if you can choose, at each point p , an equivalence class of ordered bases of $T_p M$ (two bases are equivalent if the change-of-basis matrix has positive determinant), in a way that varies continuously across M .

Three equivalent ways to package this:

1. **Pointwise:** a continuous choice of "positive frame" at each tangent space.
2. **Form-theoretic:** a nowhere-vanishing top-form $\Omega \in \Omega^n(M)$. Two such forms give the same orientation iff they differ by a positive function.
3. **Atlas-theoretic:** an atlas where every chart transition map has positive Jacobian determinant.

is the space of complex lines through the origin in $\mathbb{C}^3$, points written as $[z_0:z_1:z_2]$.
Complex dimension 2, real dimension 4.

Three charts, the standard atlas. For $k = 0,1,2$, the open set

$$U_k = \{[z_0:z_1:z_2]: z_k \neq 0\}.$$

These cover $\mathbb{CP}^2$. Local complex coordinates:

- On U_0 : $w_1 = z_1/z_0$, $w_2 = z_2/z_0$. So $[z_0:z_1:z_2] = [1:w_1:w_2]$.
- On U_1 : $\zeta_0 = z_0/z_1$, $\zeta_2 = z_2/z_1$. So $[z_0:z_1:z_2] = [\zeta_0:1:\zeta_2]$.
- On U_2 : $\eta_0 = z_0/z_2$, $\eta_1 = z_1/z_2$.

Each chart maps its open set to $\mathbb{C}^2$, which we identify with $\mathbb{R}^4$ via $w_1 = x_1 + iy_1$, $w_2 = x_2 + iy_2$. So real coordinates on U_0 are (x_1, y_1, x_2, y_2) .

Transition map $U_0 \cap U_1$. Where both $z_0, z_1 \neq 0$ — i.e., $w_1 = z_1/z_0 \neq 0$. Compute:

$$\zeta_0 = \frac{z_0}{z_1} = \frac{1}{w_1}, \zeta_2 = \frac{z_2}{z_1} = \frac{w_2}{w_1}.$$

This is a *holomorphic* map $\Phi: \{w_1 \neq 0\} \rightarrow \mathbb{C}^2$. Hold on to this — it's the heart of the calculation.

Pick a point $p \in U_0$. Its tangent space $T_p \mathbb{CP}^2$ is real 4-dimensional. The complex coordinates w_1, w_2 give the *real* coordinate frame

$$(\partial_{x_1}, \partial_{y_1}, \partial_{x_2}, \partial_{y_2}) \in T_p \mathbb{CP}^2.$$

Choose the orientation $[\partial_{x_1}, \partial_{y_1}, \partial_{x_2}, \partial_{y_2}]$.

That's a pointwise orientation at p : the equivalence class of *that* ordered basis. Why this ordering? Because it's the "complex" ordering: real, imaginary, real, imaginary. The complex structure J (multiplication by i) acts as $J \partial_{x_j} = \partial_{y_j}$ and $J \partial_{y_j} = -\partial_{x_j}$, and the chosen ordering is compatible with J in a specific sense we'll make precise in step 5.

Continuity. Vary p within U_0 : the frame $(\partial_{x_1}, \partial_{y_1}, \partial_{x_2}, \partial_{y_2})$ is the coordinate frame, smooth in p , so the chosen pointwise orientation extends continuously across U_0 .

Now extend to U_1 . Pick $q \in U_0 \cap U_1$. Two candidate frames at q :

- From U_0 : $F_0 = (\partial_{x_1}, \partial_{y_1}, \partial_{x_2}, \partial_{y_2})$ (in w -coords).
- From U_1 : $F_1 = (\partial_{u_0}, \partial_{v_0}, \partial_{u_2}, \partial_{v_2})$, where $\zeta_0 = u_0 + iv_0$, $\zeta_2 = u_2 + iv_2$.

Are these in the same equivalence class? That's a determinant computation.

Differentiating the transition Φ :

$$\zeta_0 = \frac{1}{w_1}, \zeta_2 = \frac{w_2}{w_1} \Rightarrow \frac{\partial \zeta_0}{\partial w_1} = -\frac{1}{w_1^2}, \quad \frac{\partial \zeta_0}{\partial w_2} = 0, \quad \frac{\partial \zeta_2}{\partial w_1} = -\frac{w_2}{w_1^2}, \quad \frac{\partial \zeta_2}{\partial w_2} = \frac{1}{w_1}.$$

The complex Jacobian matrix is

$$J_{\mathbb{C}} = \begin{pmatrix} -1/w_1^2 & 0 \\ -w_2/w_1^2 & 1/w_1 \end{pmatrix}, \det J_{\mathbb{C}} = -\frac{1}{w_1^3}.$$

A *complex* number, possibly negative, possibly far from positive. So you might worry. Don't! What matters is the *real* Jacobian, the 4×4 matrix of partial derivatives of (u_0, v_0, u_2, v_2) with respect to (x_1, y_1, x_2, y_2) .

Key lemma. If $f: \mathbb{C}^n \rightarrow \mathbb{C}^n$ is holomorphic with complex Jacobian $J_{\mathbb{C}}$, then its real Jacobian (as a map $\mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$) satisfies

$$\det J_{\mathbb{R}} = |\det J_{\mathbb{C}}|^2 \geq 0$$

When you integrate an n -form ω over M , you do it chart by chart and sum. In each chart, $\int_U \omega$ is computed as an ordinary Lebesgue integral of the coefficient function. If the chart transitions had negative Jacobians, two adjacent charts would disagree on the sign of the same piece of integral, and the sum would be ill-defined. Orientation is exactly the structure that prevents this.

A **manifold with boundary** M is locally modeled on the half-space $\mathbb{H}^n = \{x \in \mathbb{R}^n : x^n \geq 0\}$. The boundary ∂M is an $(n-1)$ -manifold without boundary.

The induced orientation rule. If M is oriented, the boundary inherits a canonical orientation by the **outward-normal-first rule**: a basis $(v_1, \dots, v_{n-1})$ of $T_p \partial M$ is positively oriented iff $(v, v_1, \dots, v_{n-1})$ is positively oriented in $T_p M$, where v is an outward-pointing vector. This is a convention, but it's the one that makes Stokes work without an extra sign.

1.4 Integration of forms — the mechanism

The actual definition is engineered so that the result is independent of all choices.

Step 1 — In a single oriented chart. If $\omega = f dx^1 \wedge \dots \wedge dx^n$ is supported in an oriented chart $U \subset \mathbb{R}^n$,

$$\int_M \omega := \int_U f(x) dx^1 \dots dx^n,$$

an ordinary Lebesgue integral. The ordering matches the orientation.

Step 2 — Partition of unity. For a general ω on M , pick a locally finite oriented atlas $\{U_\alpha\}$ and a partition of unity $\{\rho_\alpha\}$ subordinate to it. Then

$$\int_M \omega := \sum_{\alpha} \int_{U_\alpha} \rho_\alpha \omega.$$

Step 3 — Independence of choices. One proves this is independent of the atlas and partition by showing that on overlaps the change-of-variables formula plus the wedge-product transformation rule cancel exactly. Orientability is exactly the hypothesis that makes the cancellation produce the *same* sign in every chart.

How to think about partitions of unity. They are not a deep technical trick; they are a smooth book-keeping device. The functions ρ_α are smooth, nonnegative, locally finitely many, sum to 1, and each is supported in one chart. Multiplying any global object by ρ_α localizes it to a chart, where you can compute. The sum then reassembles a global answer. They exist on any paracompact manifold, which includes essentially every manifold you ever meet.

So far, forms do not require a metric, A smooth manifold gives you

$$T_p M, \quad T_p^* M, \quad \Omega^k(M), d, \quad \wedge, \quad F^*$$

But it does not automatically give you lengths, angles gradients, or volume. A Riemannian metric is a smoothly varying inner product:

$$g_p: T_p \times T_p \rightarrow \mathbb{R}$$

In coordinates

$$g = g_{ij} dx^i \otimes dx^j$$

The metric gives a volume form:

$$dvol g = \sqrt{\det(g_{ij})} dx^1 \wedge \dots \wedge dx^n$$

A **Riemannian metric** g on M is a smooth section of $\text{Sym}^2 T^*M$ that is positive definite at every point. Concretely: a smoothly varying inner product $g_p: T_p M \times T_p M \rightarrow \mathbb{R}$, bilinear, symmetric, positive definite.

In coordinates $(x^1, \dots, x^n)$, the metric is encoded by the matrix

$$g_{ij}(x) = g(\partial_i, \partial_j),$$

which is symmetric and positive definite at each point. The inverse matrix g^{ij} (raise indices) plays a crucial role; it's the metric on the cotangent space.

2.2 Why $\sqrt{\det g} \, dx$ is the volume element

This is the self-check from the image and the most important formula to *understand* rather than memorize.

The setup. On an n -manifold with metric g , the **Riemannian volume form** $d\text{vol}_g$ is the unique n -form such that for any positively oriented orthonormal frame $(e_1, \dots, e_n)$, $d\text{vol}_g(e_1, \dots, e_n) = 1$. This is just saying: a unit cube in g -orthonormal coordinates has volume 1, as it should.

Computation in arbitrary coordinates. In a chart, the coordinate basis $(\partial_1, \dots, \partial_n)$ is generally *not* orthonormal — the inner products are g_{ij} . To get an orthonormal frame, you do Gram–Schmidt or equivalently take the matrix square root of g . The change of basis from "orthonormal" to "coordinate" has determinant equal to $\sqrt{\det g}$. Therefore

$$d\text{vol}_g = \sqrt{\det g_{ij}} \, dx^1 \wedge \dots \wedge dx^n.$$

Quick rigorous derivation. Let A be the matrix sending the coordinate basis to an orthonormal basis: $e_a = A^i{}_a \partial_i$. Orthonormality means $\delta_{ab} = g(e_a, e_b) = A^i{}_a A^j{}_b g_{ij}$, i.e. $I = A^T G A$, so $\det A = (\det G)^{-1/2}$. The coordinate top-form on (∂_i) and the canonical top-form on (e_a) are related by $\det A$, so

$$d\text{vol}_g(\partial_1, \dots, \partial_n) = \det(A)^{-1} = \sqrt{\det G}.$$

This identifies $d\text{vol}_g$ as $\sqrt{\det g}$ times the coordinate top-form.

Definition. ∇f is the vector field metrically dual to the 1-form df . In symbols: $g(\nabla f, X) = df(X) = X(f)$ for all vector fields X .

Coordinates. $(\nabla f)^i = g^{ij} \partial_j f$. Note the inverse metric — this is where it earns its keep. In Euclidean $\mathbb{R}^n$, $g^{ij} = \delta^{ij}$ and you recover the usual gradient.

Intuition. The gradient is the direction of steepest ascent measured *with respect to the metric*. Different metrics give different gradients of the same function. On the sphere, the gradient of the latitude function points along meridians, but its magnitude depends on $\sin \theta$ through g .

2.3.2 Divergence

There are two equivalent definitions; you'll want both.

Definition A (form-theoretic, the slick one). The divergence of a vector field X is the unique function such that

$$\mathcal{L}_X(d\text{vol}_g) = (\text{div } X) d\text{vol}_g,$$

where $\mathcal{L}_X$ is the Lie derivative. In words: divergence measures the rate at which the volume form changes under the flow of X . Cartan's magic formula gives $\mathcal{L}_X\omega = d\iota_X\omega + \iota_X d\omega$, and since $d d\text{vol}_g = 0$ (top form is closed for trivial reasons),

$$(\text{div } X) d\text{vol}_g = d(\iota_X d\text{vol}_g).$$

Definition B (coordinate, the computational one).

$$\boxed{\text{div } X = \frac{1}{\sqrt{\det g}} \partial_i (\sqrt{\det g} X^i)}.$$

Why these agree. $\iota_X d\text{vol}_g$ in coordinates is the $(n-1)$ -form $\sum_i (-1)^{i-1} X^i \sqrt{\det g} dx^1 \wedge \dots \wedge \widehat{dx^i} \wedge \dots \wedge dx^n$. Apply d and divide by $\sqrt{\det g}$ and you get the coordinate formula.

Intuition. Volume distortion under the flow. If X generates a flow that pushes a small region outward, divergence is positive. The $\sqrt{\det g}$ factor is there because the volume that's being stretched is the *Riemannian* volume, not the coordinate volume.

Definition. $\Delta f = \text{div}(\nabla f)$. This is the natural generalization of the Euclidean Laplacian.

Coordinates. Combining the above:

$$\boxed{\Delta f = \frac{1}{\sqrt{\det g}} \partial_i (\sqrt{\det g} g^{ij} \partial_j f)}.$$

In Euclidean $\mathbb{R}^n$, $g^{ij} = \delta^{ij}$, $\sqrt{\det g} = 1$, and you get $\sum_i \partial_i^2 f$ as expected. On S^2 in spherical coordinates,

$$\Delta_{S^2} f = \frac{1}{\sin \theta} \partial_\theta (\sin \theta \partial_\theta f) + \frac{1}{\sin^2 \theta} \partial_\phi^2 f.$$

The $\sin \theta$ factors are exactly $\sqrt{\det g}$ and $g^{\varphi\varphi}$ appearing in their proper places. On a closed manifold (no boundary) or for compactly supported functions:

$$\int_M (\Delta f) h \, d \operatorname{vol}_g = - \int_M g(\nabla f, \nabla h) \, d \operatorname{vol}_g = \int_M f (\Delta h) \, d \operatorname{vol}_g.$$

The first equality is Green's first identity; the symmetry of the result encodes self-adjointness of Δ . With boundary, you pick up a boundary term $\int_{\partial M} h g(\nabla f, \nu) \, d\sigma$. This identity is **just Stokes applied to the $(n-1)$ -form $h \cdot \iota_{\nabla f} d \operatorname{vol}_g$** , with the Leibniz rule expanded out — see how integration of forms and Riemannian geometry are not separate subjects.